

Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: 4&5 SkillMatch Resume Matcher and Skill Recommender

Start date: 13-10-2025

Names:

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Internship Duration: 8 Weeks

1. Project Title

SkillMatch Resume Matcher and Skill Recommender

2. Project Objective

The objective of Career Align is to create an AI-based platform that supports career decision-making by analyzing resumes, identifying skills, and predicting suitable job roles. The system compares candidate profiles with job requirements, calculates match scores, and highlights areas for improvement. It provides learning recommendations, project suggestions, and visual dashboards so that both candidates and recruiters can make informed, fast, and accurate decisions.

3. Project description in detail

Career Align is an AI-driven platform that helps candidates and recruiters make better career decisions. It automatically analyzes uploaded resumes, extracts skills, and compares them with job descriptions. The system identifies matching, missing, and extra skills, and calculates a match score to show how suitable a candidate is for a job role. Based on skill patterns, it predicts relevant roles such as frontend developer, backend developer, or full stack developer. It also recommends courses and project ideas to fill skill gaps and improve job readiness. Candidates can view all results on a personal dashboard and download a report. For recruiters, the platform supports bulk resume upload, candidate comparison, and ATS-based ranking. This saves time in shortlisting and improves hiring accuracy. Overall, Career Align acts as a smart assistant for both job seekers and hiring teams.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Introduction to AI & ML Concepts	Understood AI/ML basics and finalized project objectives and scope.
Week 2	Data Collection & Preprocessing	Collected sample resumes and job descriptions and performed data preprocessing.
Week 3	Resume Parsing Module Development	Developed the resume parsing module using NLP for skill extraction.
Week 4	Job Description Analysis Module	Implemented job description analysis to identify required skills.
Week 5	Skill Recommendation Model	Designed the skill recommendation engine to suggest missing skills.
Week 6	Model Evaluation & Optimization	Optimized model performance and improved match accuracy.
Week 7	Frontend Development & Integration	Optimized model performance and improved match accuracy.
Week 8	Testing & Deployment	Performed end-to-end testing, fixed integration bugs, and deployed the complete application.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project kickoff defined objectives, scope, and roles.	13 - Oct - 2025
Prototype/First Draft	Developed the prototype for resume matching and skill recommendation.	30 - Oct - 2025
Mid-Term Review	Completed mid-term review to assess progress and refine project approach.	13 - Nov - 2025
Final Submission	Submitted the completed SkillMatch project with full functionality and documentation.	21 - Nov - 2025
Presentation	Prepared and delivered the project presentation showcasing features and outcomes.	01 - Dec - 2025

5b. Project execution details

- The work started with defining the project objectives, understanding the problem, and identifying the required functionalities.
- Real resumes and job descriptions were gathered to study how skills and role requirements are written in different formats.
- A resume processing component was built to extract information such as skills, experience, and education from uploaded files.
- Skill embeddings were generated and compared using similarity measures to calculate how closely a profile matches a job.
- A job role prediction module was created to suggest potential roles based on the combination of detected skills.
- A recommendation layer was added to guide users with missing skills, online courses, and suitable project ideas.
- Dashboards were designed to clearly show match percentage, skill breakdown, and role suggestions to users.
- Functionality for bulk resume upload and candidate comparison was implemented for the recruiter interface.
- Multiple test cases were executed using different resumes and job descriptions to check accuracy and usability.
- The final step involved preparing documentation, visuals, and presentation material to showcase the completed system.

6. Snapshots / Screenshots

Opening the project folder containing all the main files.

Name	Date modified	Type	Size
assets	23-11-2025 18:22	File folder	
components	23-11-2025 18:22	File folder	
hooks	23-11-2025 18:22	File folder	
integrations	23-11-2025 18:22	File folder	
lib	23-11-2025 18:22	File folder	
pages	23-11-2025 18:22	File folder	
services	23-11-2025 18:22	File folder	
utils	23-11-2025 18:22	File folder	
App.css	23-11-2025 18:22	CSSfile	1 KB
App	23-11-2025 18:22	TypeScript Source ...	2 KB
index.css	23-11-2025 18:22	CSSfile	14 KB
main	23-11-2025 18:22	TypeScript Source ...	1 KB
vite-env.d	23-11-2025 18:22	TS File	1 KB

Opening the *CandidateDashboard.tsx* file in VS Code to view the main CandidateDashboard application code.

```

src > pages > CandidateDashboard.tsx > ...
45  const CandidateDashboard = () => {
    // Prepare data for components
    const matchedSkills = latestAnalysis?.extracted_skills?.map((skill, idx) => ({
      name: skill,
      proficiency: "Intermediate",
      matchPercentage: 85 + Math.floor(Math.random() * 15),
    }));

    const missingSkills = latestAnalysis?.skill_gaps?.map((gap) => ({
      name: gap.skill,
      requiredLevel: "Advanced",
      priority: gap.importance === "high" ? "High" as const : gap.importance === "medium" ? "Medium" as const : "Low" as const;
    }));

    // Define diverse learning platforms with all options for each skill
    const learningPlatforms = [
      { name: "Udemy", url: "https://www.udemy.com/courses/search?q=", color: "bg-purple-600" },
      { name: "Coursera", url: "https://www.coursera.org/search?query=", color: "bg-blue-600" },
      { name: "Pluralsight", url: "https://www.pluralsight.com/search?q=", color: "bg-pink-600" },
      { name: "LinkedIn", url: "https://www.linkedin.com/learning/search?keywords=", color: "bg-blue-700" },
      { name: "edX", url: "https://www.edx.org/search?q=", color: "bg-blue-500" },
      { name: "Codecademy", url: "https://www.codecademy.com/search?query=", color: "bg-indigo-600" },
      { name: "FreeCodeCamp", url: "https://www.freecodecamp.org/news/search/?query=", color: "bg-green-600" },
      { name: "YouTube", url: "https://www.youtube.com/results?search_query=", color: "bg-red-600" },
    ];

    const skillRecommendations = latestAnalysis?.skill_gaps?.slice(0, 5).map((gap) => {
      // Generate all platform resources for this skill
      const platformResources = learningPlatforms.map(platform => ({
        name: platform.name,
        url: `${platform.url}${gap.skill}`,
        type: "course" as const,
        color: platform.color,
      }));
    });
  }

```

Opening the *HRDashboard.tsx* file in VS Code to view the main HRDashboard application code.

```

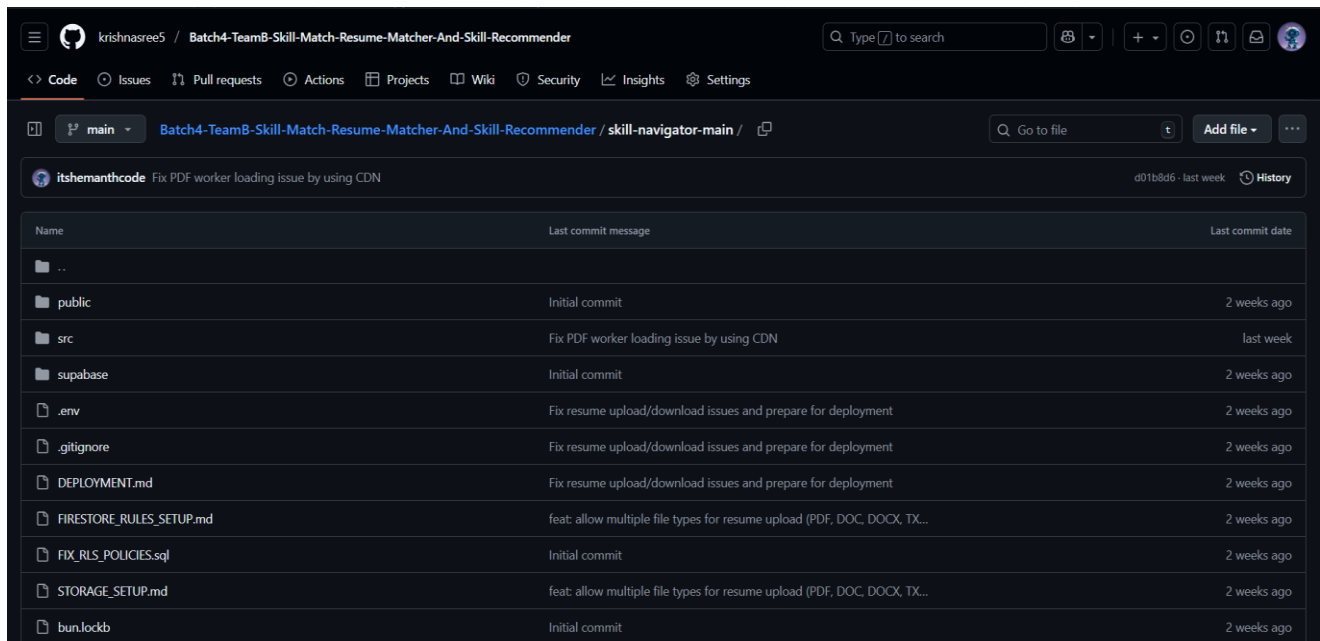
src > pages > HRDashboard.tsx > ...
61  const HRDashboard = () => {
    const filteredResumes = resumes
      .filter(resume => {
        const matchesScore =
          (resume.matchScore || 0) >= matchScoreRange[0] &&
          (resume.matchScore || 0) <= matchScoreRange[1];
        return matchesSearch && matchesScore;
      })
      .sort((a, b) => {
        if (sortBy === "score") {
          return (b.matchScore || 0) - (a.matchScore || 0);
        } else if (sortBy === "date") {
          return new Date(b.uploaded_at).getTime() - new Date(a.uploaded_at).getTime();
        } else {
          return (a.file_name || "").localeCompare(b.file_name || "");
        }
      })
      .slice(0, topCandidatesCount);

    const handleAnalyzeResumes = async () => {
      if (!jobDescription.trim()) {
        toast({
          title: "Job description required",
          description: "Please enter a job description before analyzing resumes",
          variant: "destructive",
        });
        return;
      }

      if (resumes.length === 0) {
        toast({
          title: "No resumes found",
          description: "Please upload resumes before analyzing",
          variant: "destructive",
        });
        return;
      }
    };
  }

```

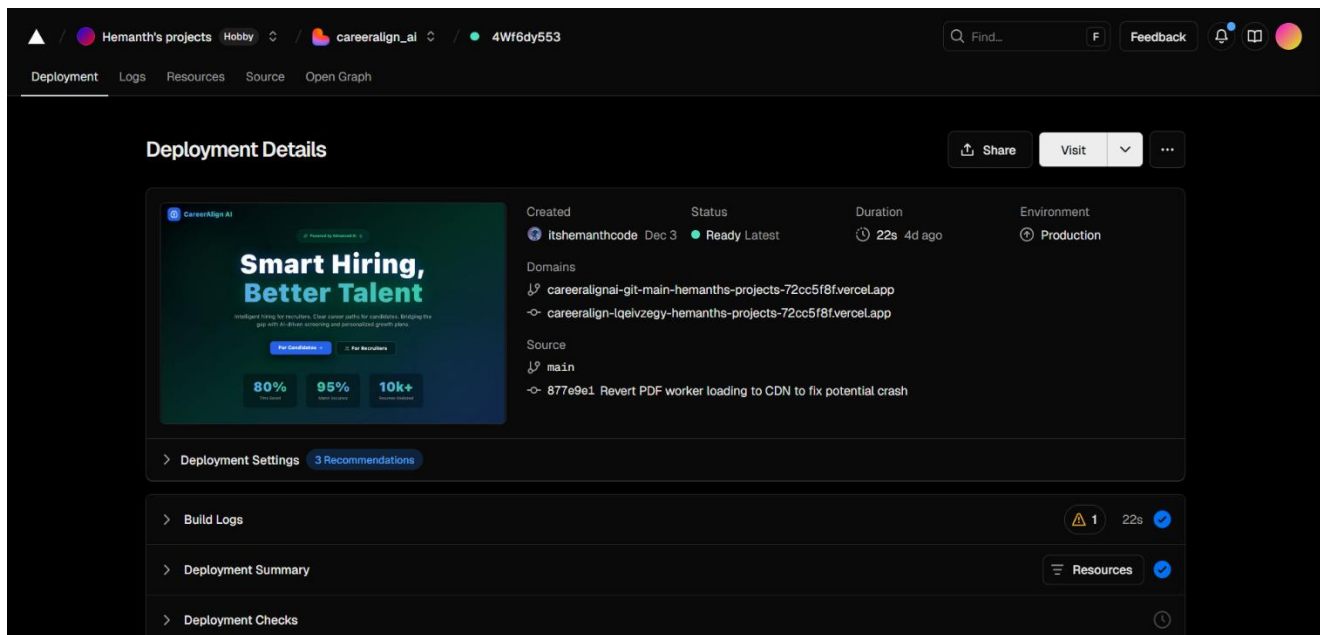
Pushed the entire project into GitHub



The screenshot shows a GitHub repository page for 'Batch4-Team8-Skill-Match-Resume-Matcher-And-Skill-Recommender'. The repository is owned by 'krishnasree5'. The main branch is 'main'. The repository description is 'Fix PDF worker loading issue by using CDN'. The commit history table is as follows:

Name	Last commit message	Last commit date
..		
public	Initial commit	2 weeks ago
src	Fix PDF worker loading issue by using CDN	last week
supabase	Initial commit	2 weeks ago
.env	Fix resume upload/download issues and prepare for deployment	2 weeks ago
.gitignore	Fix resume upload/download issues and prepare for deployment	2 weeks ago
DEPLOYMENT.md	Fix resume upload/download issues and prepare for deployment	2 weeks ago
FIRESTORE_RULES_SETUP.md	feat: allow multiple file types for resume upload (PDF, DOC, DOCX, TX...	2 weeks ago
FIX_RLS_POLICIES.sql	Initial commit	2 weeks ago
STORAGE_SETUP.md	feat: allow multiple file types for resume upload (PDF, DOC, DOCX, TX...	2 weeks ago
bun.lockb	Initial commit	2 weeks ago

Deployed the project Successfully on Vercel (<https://careeralign>).



The screenshot shows a Vercel deployment page for 'careeralign_ai'. The deployment is titled 'Smart Hiring, Better Talent'. The deployment details are as follows:

Created	Status	Duration	Environment
Itshemanthcode Dec 3	Ready Latest	22s 4d ago	Production

Domains:

- careeralignai-git-main-hemanths-projects-72cc5f8f.vercel.app
- careeralign-lqelvzgy-hemanths-projects-72cc5f8f.vercel.app

Source:

- main
- 877e9e1 Revert PDF worker loading to CDN to fix potential crash

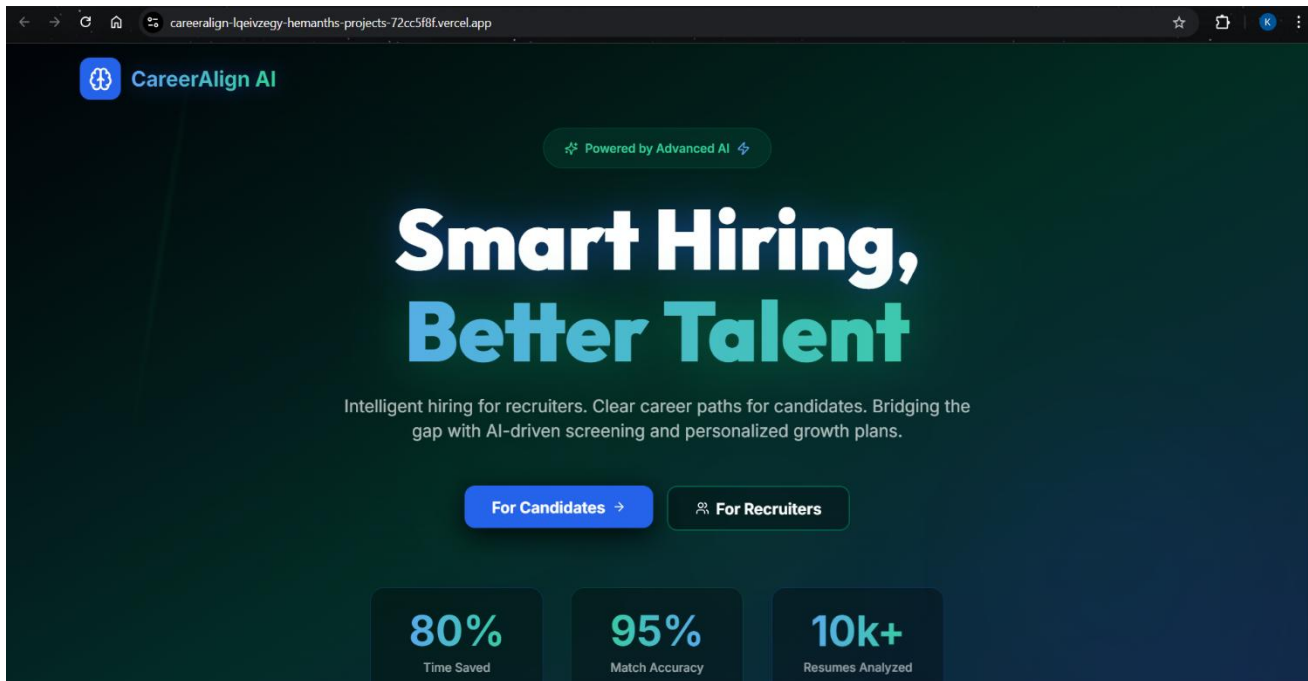
Deployment Settings: 3 Recommendations

Build Logs: 1 22s

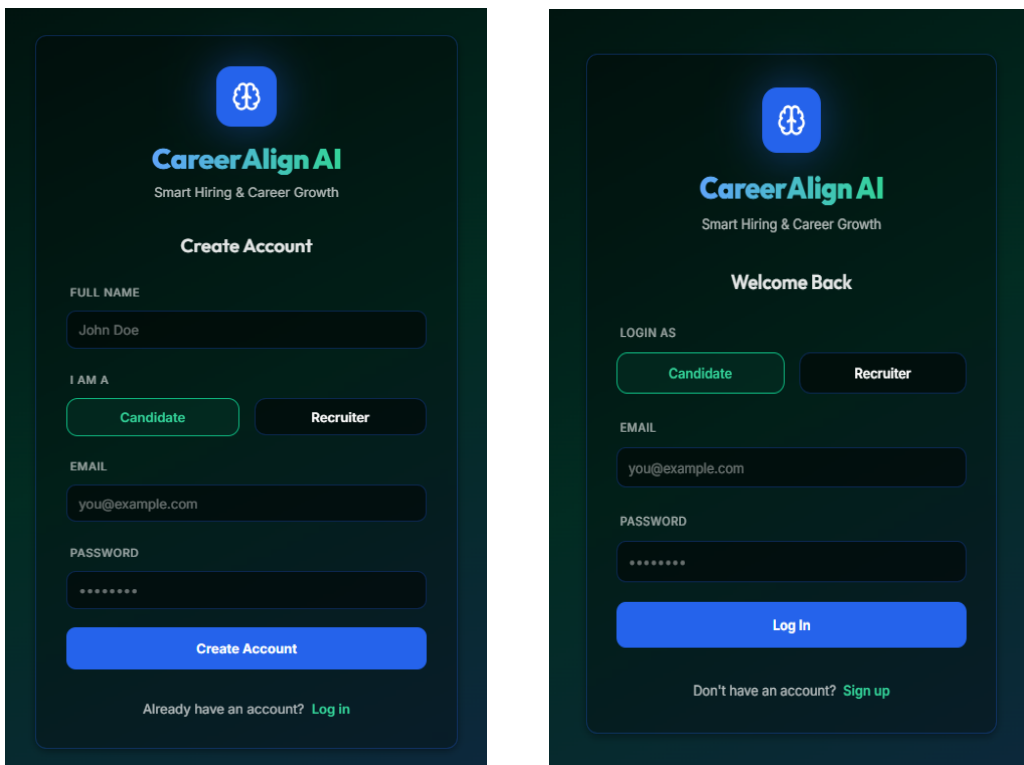
Deployment Summary: Resources

Deployment Checks: 1

The Landing page of the CareerAlign web application appears on the browser.



The Signup and Login pages of CareerAlign(Filling in user details such as name, user type, email and password in the registration form).



Submitting the registration form — message displayed: ***Registered successfully.***

The CandidateDashboard page of the CareerAlign web application appears on the browser.

Candidate Dashboard
Track your skills, get insights, and grow your career

Upload Your Resume
PDF, DOC, DOCX, TXT, RTF format, max 10MB
Choose File

Resume History
Select a previously uploaded resume to analyze
Select a resume
Delete Selected

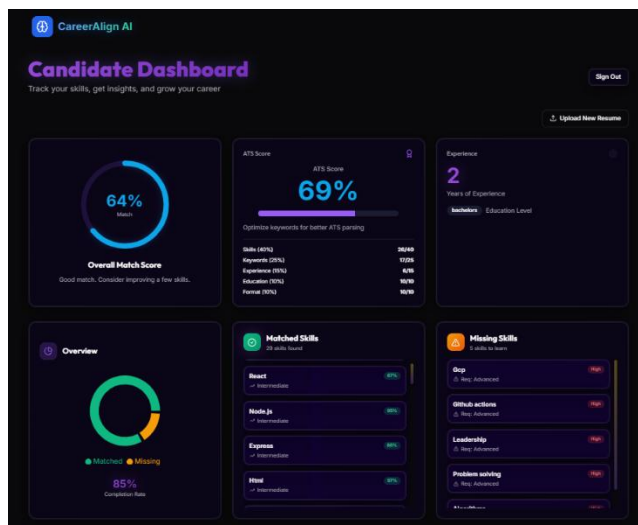
Job Role Selection
Select a job role or paste a custom job description
Select job role...
Or paste custom Job Description:
Example: We're looking for a React developer with 3+ years experience, strong TypeScript, testing, and API integration skills...
65000 characters
Analyze Match

Candidate Dashboard
Track your skills, get insights, and grow your career

Upload Your Resume
PDF, DOC, DOCX, TXT, RTF format, max 10MB
Choose File

Resume History
Select a previously uploaded resume to analyze
Kuruba Hemant Kishore.pdf (06/12/2025)
Delete Selected

Job Role Selection
Select a job role or paste a custom job description
Select job role...
Or paste custom Job Description:
Job Title: Software Engineer - Fresher
Location: (City, State) / Hybrid / Remote
Employment Type: Full-time
About the Role:
We are looking for enthusiastic and passionate fresh graduates to join our software development team. As a Software Engineer (Fresher), you will work closely with senior developers to build high-quality software solutions.
65000 characters
Analyzing...



Candidate Dashboard
Track your skills, get insights, and grow your career

Trending Skills
Boost your profile with these in-demand skills

Projects to Boost Your Resume
Build these projects to close skill gaps

Build a Full-Stack Web Application
Create an end-to-end application with authentication, database, and modern UI
React, Node.js, PostgreSQL, REST API
Intermediate
2-4 weeks
Impact: 77%

Containerize and Deploy Application
Deploy your application using Docker and cloud services
Docker, AWS/Azure, CI/CD, DevOps
Intermediate
2-4 weeks
Impact: 78%

Open Source Contribution
Contribute to popular open-source projects to build community presence
Git, GitHub, Collaboration, Code Review
Intermediate
2-4 weeks
Impact: 77%

Action Items & Next Steps
1. Focus on Missing Skills
2. Complete Recommended Projects

The HR-Dashboard page of the CareerAlign web application appears on the browser.

HR Dashboard
Manage candidates and streamline your recruitment process

Next Resumes 3
Shortlisted 0
Auto-check Score 47%
Pending Review 0

Bulk Upload Resumes
Upload multiple candidate resumes for AI-powered analysis and ranking

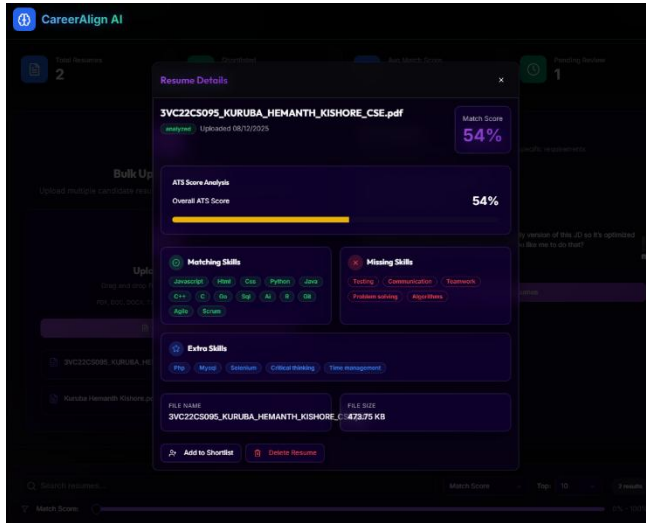
Upload Resumes
Drag and drop files here, or click to browse
PDF, DOC, DOCX, TXT, RTF format, max 10MB per file
Choose File

Job Description
Add a job description to match candidates against specific requirements
- Effectively articulate technical challenges and solutions.
- Adapt to handling ambiguous or undefined problems as well as ability to think abstractly.
Our inclusive culture empowers Amazonians to deliver the best results for our customers. If you have a disability and need a workplace accommodation or adjustment during the application and hiring process, including support for the interview or onboarding process, please visit https://amazon.jobs/content/en/how-we-hire/accommodations for more information. If you're having trouble applying in our system, please contact your Recruiting Partner.

Upload Resumes
Drag and drop files here, or click to browse
PDF, DOC, DOCX, TXT, RTF format, max 10MB per file
Choose File

Top Candidates
Export CSV | Analytics

Rank	Candidate Name	Match Score	Status	Uploaded	Actions
#1	KrishnaDinesh_CoinSwitch.pdf	50%	Analysed	12/1/2025	🗑️ 🔄 📄
#2	KrishnaDinesh_Amazon_277102025.pdf	50%	Analysed	12/1/2025	🗑️ 🔄 📄
#3	KrishnaDinesh_BITCS.pdf	42%	Analysed	12/1/2025	🗑️ 🔄 📄



	A	B	C	D	E	F	G	H	I	J	K
1	Name	Match Score	Skills	Experience	Education	Status					
2	3VC22CS095_KURUBA_HEMANTH_KISHORE_CSE.pdf	54%	08-Dec	5+ years	BS in CS	analyzed					
3	Kuruba Hemant Kishore	39%	08-Dec	5+ years	BS in CS	analyzed					
4											
5											
6											
7											
8											
9											

7. Challenges Faced

- **Unstructured Resume Content:** Many resumes contained tables, bullet points, and images, which made it difficult to extract information cleanly without losing important text.
- **Non-Technical Skill Identification:** Soft skills such as teamwork, leadership, and communication were difficult to detect automatically because they were not listed like technical skills.
- **Multiple Job Role Confusion:** Some candidates had experience across different areas, which made it challenging to decide the most suitable job role from their skill combinations.
- **Threshold Selection for Similarity:** Choosing the right similarity threshold was tricky. A low threshold produced many unrelated matches, while a high threshold ignored meaningful matches.
- **Handling Synonyms and Abbreviations:** Terms like “ML” and “Machine Learning” or “JS” and “JavaScript” had to be treated as the same skill, which required additional mapping logic.
- **Scalability Issues:** As the number of resumes increased, processing time and memory usage increased, requiring performance optimization in the backend.
- **Real-Time Recommendations:** Suggesting courses and projects dynamically based on skill gaps required reliable sources and filtering mechanisms to avoid irrelevant suggestions.
- **User Interface Consistency:** Designing dashboards that were simple for both students and HR users, without overwhelming them with too much data, was a UX challenge.

8. Learnings & Skills Acquired

- **End-to-End System Development:** Gained experience in building a complete AI platform from data input (resume parsing) to final output (job prediction and recommendations).
- **AI for HR and Career Development:** Applied AI concepts like skill extraction, semantic similarity, and job role prediction to enhance talent discovery and career evaluation.
- **Dual-Role Platform Design:** Designed distinct yet integrated features and dashboards tailored for two user types: Candidates and HR/Recruiters.
- **Software Architecture:** Integrated frontend and backend technologies to create a scalable, efficient, and robust system.
- **Resume Parsing and Text Extraction:** Implemented resume upload and PDF parsing to extract text from documents.
- **Natural Language Processing (NLP) & Embeddings:** Utilized the Xenova/all-MiniLM-L6-v2 transformer model to convert skills into vector embeddings. Applied embeddings to calculate semantic similarity using cosine similarity for accurate skill matching.
- **AI-Driven Analysis:** Developed algorithms for automatic skill extraction and evaluation. Implemented ATS score calculation for candidates.
- **Matching and Recommendation Systems:** Mastered skill matching strategies combining exact and semantic matches (threshold > 0.65) to reduce false positives. Developed a job role prediction system based on detected skill combinations. Designed modules for skill gap analysis and personalized learning/project suggestions.
- **Dashboard Development:** Built sophisticated user interfaces displaying complex data, including match score display, skills breakdown visualization, and analytics dashboard.
- **HR-Specific Functionality:** Created tools for recruiters such as bulk resume upload, candidate comparison, and ATS score ranking.

9. Testimonials from team

- This project enhanced our understanding of machine learning models for text similarity and helped us learn how to extract meaningful insights from unstructured data.
- We gained hands-on experience in resume parsing, skill recommendation systems, and building interactive dashboards, which strengthened both my technical and practical skills."
- "CareerAlign" taught us how to work collaboratively, manage project timelines, and integrate AI solutions into a functional web application."
- The project improved our knowledge in NLP, data preprocessing, and visualization techniques, and also boosted my confidence in handling real-world AI applications.

10. Conclusion:

CareerAlign AI stands as a comprehensive, AI-driven platform that fundamentally enhances talent acquisition and career development. The system's architecture supports a dual mandate, providing advanced tools for both Candidates and HR/Recruiters.

The platform's core capability lies in its advanced processes, including end-to-end AI resume analysis, accurate skill extraction, ATS score calculation, and sophisticated job role prediction. Its Skill Matching feature leverages semantic similarity, employing the Xenova/all-MiniLM-L6-v2 transformer model to convert skills into vector embeddings and calculate precise match scores via cosine similarity.

By integrating features like Skill Gap Analysis and personalized learning recommendations, CareerAlign AI delivers more than just evaluation; it fosters growth. This efficient, scalable solution, built on a cutting-edge technology stack, positions it as a pivotal tool for driving the future of smart hiring and informed talent growth.

11.Acknowledgements

We would like to express our heartfelt gratitude to Infosys Springboard for giving us the opportunity to work on the SkillMatch: Resume Matcher and Skill Recommender project. This internship provided us with an enriching environment to explore AI-driven resume analysis, skill intelligence, and machine learning applications in modern recruitment systems.

We are especially grateful to our mentor, Sangeetha Mahalingam, for her continuous guidance, insightful feedback, and unwavering support throughout the project. Her mentorship helped us navigate challenges, refine our understanding of complex AI concepts, and grow both technically and professionally.

We would also like to extend our sincere thanks to our fellow team members for their collaboration, encouragement, and collective efforts in building and enhancing the project's functionalities. The teamwork, shared ideas, and coordinated efforts made this journey smooth, productive, and truly enriching.

Finally, we appreciate everyone who supported us directly or indirectly in our learning and in the successful completion of this project. This experience has strengthened our technical skills and broadened our perspective on AI-based talent solutions, making this internship truly memorable and transformative.